YEAR 10 MATHS · CHAPTER 3 · MIND MAP

Indices, exponentials and logarithms

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1 Review of index laws

$a^m a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$

$(2x^3)^2 \times x = 4x^6 \times x = 4x^7$

2 Negative indices

$a^{-n} = \frac{1}{a^n}$

$3^{-2} = \frac{1}{9}$ $2x^{-3} = \frac{2}{x^3}$

$\left(\frac{2}{3}\right)^{-2} = \frac{9}{4}$

Watch out: only the x is to the power -3 in $2x^{-3}$.

3 Scientific notation

$a \times 10^n$ with $1 \le a < 10$.

$0.000\,52 = 5.2 \times 10^{-4}$

$(3 \times 10^5)(4 \times 10^{-2}) = 12 \times 10^3 = 1.2 \times 10^4$

4 Fractional indices

$a^{\frac{1}{n}} = \sqrt[n]{a}$ and $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$

$27^{\frac{2}{3}} = 3^2 = 9$ $16^{-\frac{1}{2}} = \frac{1}{4}$

5 Exponential equations

Write both sides with the same base, then equate the indices.

$2^x = 32 = 2^5 \Rightarrow x = 5$

$9^x = 27 \Rightarrow 3^{2x} = 3^3 \Rightarrow x = \frac{3}{2}$

6 Exponential graphs

$y = a^x$ $(a > 1)$ passes through $(0, 1)$ and gets close to $y = 0$ but never reaches it.

$y = 2^x$: $(0, 1), (1, 2), (2, 4)$

$y = 2^{-x}$ is its reflection in the y-axis

7 Exponential growth and decay

Growth: $A = A_0(1 + r)^n$. Decay: $A = A_0(1 - r)^n$.

2000 people growing 5% a year: $2000 \times 1.05^3 \approx 2315$ after 3 years

Decay 10% a year: multiply by 0.9 each year

8 Compound interest

$A = P\left(1 + \frac{r}{k}\right)^{nk}$: k payments a year for n years.

$5000 at 6% p.a. compounded monthly for 2 years: $5000 \times 1.005^{24} \approx \$5635.80$

Watch out: divide the yearly rate by 12 for monthly.

9 Introducing logarithms

$\log_a b = x$ means $a^x = b$.

$\log_2 8 = 3$ $\log_{10} 1000 = 3$ $\log_3 \frac{1}{9} = -2$

10 Logarithmic scales

Each step up the scale is 10 times bigger.

A magnitude 6 earthquake has 10 times the amplitude of a magnitude 5

pH 3 is 10 times more acidic than pH 4

11 Laws of logarithms

$\log xy = \log x + \log y$, $\log \frac{x}{y} = \log x - \log y$, $\log x^n = n \log x$

$\log_2 12 - \log_2 3 = \log_2 4 = 2$

$\log_a 1 = 0$ $\log_a a = 1$

12 Solving with logarithms

Take logs of both sides when you can't match the bases.

$3^x = 20 \Rightarrow x = \frac{\log 20}{\log 3} \approx 2.73$

Check: $3^{2.73} \approx 20$ ✓